Dump's Outline: Grade 07 - 08

1. **What is Machine Learning?**
   - Basic Concepts Explained
2. **Basic Concepts**
   - Data
   - Algorithms
   - Model
3. **Types of Machine Learning**
   - Supervised and Unsupervised Learning
4. **How Machine Learning Works?**
   - LifeCycle of Machine Learning
5. **ML Real-World Examples**
   - Machine Learning in Voice Assistants and Search Engines
6. **Solving Problems with ML**
   - Apply ML Concepts to Real-World Scenarios

Dumps on Machine Learning

1. **What does Machine Learning do?**

   A) It helps computers learn from data

   B) It makes food

   C) It plays games

   D) It reads books

   Answer: A) It helps computers learn from data

2. **What is an example of Machine Learning in real life?**

   A) Playing chess

   B) Recommending shows on Netflix

   C) Reading emails

   D) Watching TV

   Answer: B) Recommending shows on Netflix

3. **What is data in Machine Learning?**

   A) A type of game

   B) Information used to train models

   C) A movie

   D) A book

   Answer: B) Information used to train models

4. **What is an algorithm in Machine Learning?**

   A) A type of computer

   B) A set of rules to guide learning

   C) A person who programs computers

D) A kind of game

Answer: B) A set of rules to guide learning

5. **What do recommendation systems do?**

A) Suggest things you might like

B) Print documents

C) Cook food

D) Play music

Answer: A) Suggest things you might like

6. **What is a spam filter used for?**

A) To find lost emails

B) To block junk emails

C) To organize emails

D) To delete emails

Answer: B) To block junk emails

7. **How does Netflix use Machine Learning?**

A) To suggest shows based on what you've watched

B) To store movies

C) To design posters

D) To make snacks

Answer: A) To suggest shows based on what you've watched

8. **What does supervised learning involve?**

A) Learning from labeled data

B) Learning without guidance

C) Learning from random data

D) Learning from games

Answer: A) Learning from labeled data

9. What happens when a machine learns from data?

A) It forgets everything

B) It creates a model to make predictions

C) It stops working

D) It plays a game

Answer: B) It creates a model to make predictions

10. What is unsupervised learning?

A) Learning with guidance

B) Learning with no labeled data

C) Learning to play games

D) Learning from movies

Answer: B) Learning with no labeled data

11. What is a model in Machine Learning?

A) A person who builds computers

B) A set of rules to make predictions

C) A video game character

D) A type of algorithm

Answer: B) A set of rules to make predictions

12. What is the first step in Machine Learning?

A) Testing the model

B) Data collection

C) Playing games

D) Writing code

Answer: B) Data collection

13. What happens after a model is trained with data?

A) It is deleted

B) It is tested on new data

C) It plays games

D) It stops working

Answer: B) It is tested on new data

14. How can Machine Learning help doctors?

A) By making appointments

B) By analyzing patient data to suggest treatments

C) By writing prescriptions

D) By making them coffee

Answer: B) By analyzing patient data to suggest treatments

15. What type of data is used in supervised learning?

A) Labeled data

B) Unlabeled data

C) Random data

D) Deleted data

Answer: A) Labeled data

16. How do search engines use Machine Learning?

A) By providing the best search results based on user data

B) By playing music

C) By making advertisements

D) By organizing photos

Answer: A) By providing the best search results based on user data

17. What is the purpose of a spam filter in email?

A) To organize emails

B) To block unwanted emails

C) To send messages to friends

D) To create emails

Answer: B) To block unwanted emails

18. What is data preparation in Machine Learning?

A) Cleaning and organizing the data

B) Writing code

C) Collecting random data

D) Making graphs

Answer: A) Cleaning and organizing the data

19. What is an example of Machine Learning used in traffic management?

A) Counting cars manually

B) Analyzing traffic data to improve traffic flow

C) Playing music in the car

D) Designing new roads

Answer: B) Analyzing traffic data to improve traffic flow

20. How does a voice assistant improve over time?

A) By getting updates

B) By learning your speech patterns and preferences

C) By changing its name

D) By taking breaks

Answer: B) By learning your speech patterns and preferences

21. What is Machine Learning?

A) A game

B) A way for computers to learn from data

C) A type of car

D) A kind of music

Answer: B) A way for computers to learn from data

22. How can Machine Learning help with healthcare?

A) By playing doctor games

B) By analyzing patient data to detect diseases

C) By designing hospital buildings

D) By making doctor appointments

Answer: B) By analyzing patient data to detect diseases

23. What is an example of unsupervised learning?

A) Grouping animals based on their features

B) Teaching a computer to recognize cats and dogs

C) Writing a story

D) Learning from labeled data

Answer: A) Grouping animals based on their features

24. What does a voice assistant use Machine Learning for?

A) To understand speech and respond accurately

B) To clean the house

C) To send emails

D) To play games

Answer: A) To understand speech and respond accurately

25. What is an algorithm used in Machine Learning?

A) To provide entertainment

B) To help the machine learn from data

C) To make music

D) To organize books

Answer: B) To help the machine learn from data

26. What does the term "model" refer to in Machine Learning?

A) A set of predictions

B) A video game character

C) A set of rules for making predictions

D) A type of data

Answer: C) A set of rules for making predictions

27. What do supervised learning models require?

A) Labeled data

B) Random data

C) Games

D) Unlabeled data

Answer: A) Labeled data

28. What happens after training a machine learning model?

A) It becomes a game

B) It makes predictions on new data

C) It forgets everything

D) It plays music

Answer: B) It makes predictions on new data

29. What does a model do in Machine Learning?

A) It stores data

B) It makes predictions about new data

C) It plays games

D) It organizes photos

Answer: B) It makes predictions about new data

30. What is the purpose of deploying a Machine Learning model?

A) To make predictions in real-world situations

B) To teach the computer how to play chess

C) To make the model learn more data

D) To create videos

Answer: A) To make predictions in real-world situations

31. How does a machine learn from data?

A) By using an algorithm to find patterns

B) By playing games

C) By reading books

D) By watching movies

Answer: A) By using an algorithm to find patterns

32. How does Machine Learning help in traffic management?

A) By analyzing traffic data

B) By cleaning the streets

C) By designing traffic signs

D) By making roads wider

Answer: A) By analyzing traffic data

33. What is a real-world example of Machine Learning in action?

A) Sorting out spam emails

B) Cooking dinner

C) Playing video games

D) Painting pictures

Answer: A) Sorting out spam emails

34. What is the goal of data collection in Machine Learning?

A) To collect as much random data as possible

B) To gather useful data for learning

C) To play games

D) To watch TV

Answer: B) To gather useful data for learning

35. What is an example of unsupervised learning?

A) Grouping fruits by color

B) Recognizing cats and dogs in pictures

C) Learning to play chess

D) Organizing photos in an album

Answer: A) Grouping fruits by color

36. What is the role of a model in Machine Learning?

A) It helps make predictions or decisions

B) It is a game character

C) It organizes data

D) It plays music

Answer: A) It helps make predictions or decisions

37. How does a machine learn from data?

A) By looking for patterns with an algorithm

B) By playing games

C) By reading books

D) By drawing pictures

Answer: A) By looking for patterns with an algorithm

38. What happens when a model is deployed?

A) It plays games

B) It is used to make real-world decisions

C) It goes to sleep

D) It plays music

Answer: B) It is used to make real-world decisions

39. How do voice assistants get better over time?

A) By learning from your speech patterns

B) By listening to music

C) By playing games

D) By reading books

Answer: A) By learning from your speech patterns

40. What does unsupervised learning teach the machine?

A) To recognize patterns without labels

B) To follow strict rules

C) To play games

D) To watch TV

Answer: A) To recognize patterns without labels

41. What is the role of a spam filter in email systems?

A) To organize emails

B) To block unwanted or junk emails

C) To send emails to friends

D) To write emails

Answer: B) To block unwanted or junk emails

42. What type of data is used in unsupervised learning?

A) Labeled data

B) Random data

C) Unlabeled data

D) Predefined data

Answer: C) Unlabeled data

43. How do recommendation systems in apps work?

A) By guessing what you might like based on past choices

B) By choosing random options

C) By following instructions

D) By asking your friends

Answer: A) By guessing what you might like based on past choices

44. What is an example of a device that uses Machine Learning to help with navigation?

A) Smartphone GPS

B) Television

C) Refrigerator

D) Microwave

Answer: A) Smartphone GPS

45. What is a key feature of unsupervised learning?

A) It uses labeled data

B) It does not require labels for the data

C) It makes random guesses

D) It works without any data

Answer: B) It does not require labels for the data

46. How can Machine Learning help identify objects in photos?

A) By recognizing patterns in the image

B) By guessing the colors of objects

C) By reading the text in the image

D) By drawing the object

Answer: A) By recognizing patterns in the image

47. What does it mean when a machine "learns" something?

A) It memorizes all data

B) It identifies patterns from the data

C) It stops working

D) It deletes the data

Answer: B) It identifies patterns from the data

48. What is a feature in Machine Learning?

A) A person who helps the machine

B) A characteristic or property of the data

C) A tool to write code

D) A type of model

Answer: B) A characteristic or property of the data

49. How do search engines use Machine Learning?

A) By organizing photos

B) By providing the best search results based on user queries

C) By making calls

D) By sending emails

Answer: B) By providing the best search results based on user queries

50. What is a machine learning "model"?

A) A toy

B) A set of rules or algorithms for making predictions

C) A type of robot

D) A program for writing stories

Answer: B) A set of rules or algorithms for making predictions

51. How does a recommendation system like YouTube work?

A) It guesses what videos you might like based on your previous choices

B) It suggests random videos

C) It watches videos for you

D) It deletes your videos

Answer: A) It guesses what videos you might like based on your previous choices

52. What is an example of Machine Learning being used in healthcare?

A) Diagnosing diseases from medical images

B) Cooking food

C) Cleaning the hospital

D) Making appointments

Answer: A) Diagnosing diseases from medical images

53. How does a movie recommendation system work on Netflix?

A) It suggests movies based on the movies you have watched before

B) It chooses random movies for you

C) It asks your friends for suggestions

D) It plays the same movie every time

Answer: A) It suggests movies based on the movies you have watched before

54. What type of learning involves both input data and labels to train the model?

A) Supervised learning

B) Unsupervised learning

C) Reinforcement learning

D) Random learning

Answer: A) Supervised learning

55. What is the purpose of data in Machine Learning?

A) To train the model

B) To create games

C) To build websites

D) To design movies

Answer: A) To train the model

56. How does a machine "test" its learning?

A) By playing games

B) By using new data to make predictions

C) By deleting data

D) By watching videos

Answer: B) By using new data to make predictions

57. How does Machine Learning help in detecting fraud in banks?

A) By analyzing patterns in transactions

B) By organizing files

C) By printing checks

D) By calling customers

Answer: A) By analyzing patterns in transactions

58. What happens when a model is trained with enough data?

A) It makes predictions or decisions based on new data

B) It forgets everything

C) It stops working

D) It deletes the data

Answer: A) It makes predictions or decisions based on new data

59. How does Machine Learning help in face recognition?

A) By analyzing features like the eyes, nose, and mouth

B) By guessing what the face looks like

C) By drawing the face

D) By asking people to identify the face

Answer: A) By analyzing features like the eyes, nose, and mouth

60. How does a self-driving car use Machine Learning?

A) By learning to drive by observing and analyzing its surroundings

B) By driving itself without any data

C) By following a map

D) By making random turns

Answer: A) By learning to drive by observing and analyzing its surroundings

61. What is the difference between supervised and unsupervised learning?

A) Supervised learning uses labeled data, unsupervised does not

B) Supervised learning is faster

C) Unsupervised learning does not require any data

D) Unsupervised learning uses games to learn

Answer: A) Supervised learning uses labeled data, unsupervised does not

62. What is the role of a feature in Machine Learning?

A) It is a characteristic or property of the data used for learning

B) It is a game in which the machine learns

C) It is a part of the code

D) It is a tool to organize data

Answer: A) It is a characteristic or property of the data used for learning

63. What do self-driving cars use Machine Learning for?

A) To drive themselves by learning from data

B) To stop at red lights

C) To play games

D) To organize their interior

Answer: A) To drive themselves by learning from data

64. What is a recommendation system?

A) A system that suggests things based on your preferences or past actions

B) A system that plays music

C) A system that organizes files

D) A system that creates documents

Answer: A) A system that suggests things based on your preferences or past actions

65. How do social media platforms use Machine Learning?

A) By recommending posts and friends based on your interactions

B) By playing music

C) By organizing photos

D) By designing the website

Answer: A) By recommending posts and friends based on your interactions

66. How does Machine Learning help with language translation?

A) By analyzing sentence structure and context to translate languages

B) By guessing the words

C) By using a dictionary

D) By reading books

Answer: A) By analyzing sentence structure and context to translate languages

67. What is a "training dataset"?

A) Data used to train the model

B) Data used to play games

C) Data used to make movies

D) Data that is deleted after use

Answer: A) Data used to train the model

68. What is the goal of supervised learning?

A) To make predictions based on labeled data

B) To guess random answers

C) To play games with the data

D) To delete unwanted data

Answer: A) To make predictions based on labeled data

69. How does Machine Learning help in voice recognition?

A) By learning the sounds and words spoken to understand speech

B) By guessing the words

C) By reading text aloud

D) By making random noises

Answer: A) By learning the sounds and words spoken to understand speech

70. How do machines make predictions after learning?

A) By using the patterns they learned from data

B) By guessing randomly

C) By ignoring the data

D) By reading books

Answer: A) By using the patterns they learned from data

71. What does "training" a model mean in Machine Learning?

A) Teaching the model to recognize patterns in data

B) Playing games with the model

C) Writing code

D) Storing data in a database

Answer: A) Teaching the model to recognize patterns in data

72. How can Machine Learning help in shopping websites?

A) By recommending products based on your browsing history

B) By organizing the website design

C) By creating discount coupons

D) By making phone calls

Answer: A) By recommending products based on your browsing history

73. How do machines improve over time in Machine Learning?

A) By learning from more data and making better predictions

B) By stopping to learn

C) By deleting old data

D) By ignoring new data

Answer: A) By learning from more data and making better predictions

74. What type of data does unsupervised learning work with?

A) Labeled data

B) Random data

C) Unlabeled data

D) Predefined data

Answer: C) Unlabeled data

75. What is one way Machine Learning is used in healthcare?

A) Diagnosing diseases from medical images

B) Cooking food

C) Playing games

D) Cleaning the hospital

Answer: A) Diagnosing diseases from medical images

76. What is Artificial Intelligence (AI)?

A) A way to make machines think like humans

B) A type of human brain

C) A type of software

D) A type of hardware

Answer: A) A way to make machines think like humans

77. What does a chatbot do?

A) It plays music

B) It talks with people and answers questions

C) It makes phone calls

D) It sends emails automatically

Answer: B) It talks with people and answers questions

78. Which of the following is a feature of robots that use AI?

A) They can work without power

B) They can learn and make decisions on their own

C) They can speak every language

D) They can sleep

Answer: B) They can learn and make decisions on their own

79. What is an example of an AI-based system in smartphones?

A) Camera apps that recognize faces

B) Game apps for fun

C) Music players

D) Weather apps

Answer: A) Camera apps that recognize faces

80. Which device is a good example of AI used in daily life?

A) Smart speakers that can answer questions

B) A pencil

C) A TV remote

D) A clock

Answer: A) Smart speakers that can answer questions

81. How does voice recognition software in phones work?

A) It listens to your voice and converts it into text or actions

B) It plays music when you speak

C) It makes phone calls automatically

D) It types without you speaking

Answer: A) It listens to your voice and converts it into text or actions

82. What does it mean when AI is used for "predicting weather"?

A) It guesses the weather randomly

B) It analyzes weather patterns from past data to predict future weather

C) It stops predicting after some time

D) It uses a weather map

Answer: B) It analyzes weather patterns from past data to predict future weather

83. How does AI help in video streaming services like Netflix?

A) By suggesting shows and movies based on your watching history

B) By playing random shows

C) By turning off the screen

D) By creating random ads

Answer: A) By suggesting shows and movies based on your watching history

84. What is an example of AI in healthcare?

A) Robots that help in surgeries

B) TVs that show medical information

C) Beds that adjust their positions automatically

D) Computers that make appointments

Answer: A) Robots that help in surgeries

85. What is an "algorithm"?

A) A set of instructions for solving problems

B) A tool for drawing

C) A type of computer hardware

D) A kind of music

Answer: A) A set of instructions for solving problems

86. How do self-driving cars use AI?

A) By learning from data to make decisions and drive safely

B) By following a fixed path

C) By stopping at every intersection

D) By asking for directions from people

Answer: A) By learning from data to make decisions and drive safely

87. How does AI help in playing games?

A) By controlling the game characters and making decisions based on data

B) By guessing the player's moves

C) By creating random challenges

D) By playing without rules

Answer: A) By controlling the game characters and making decisions based on data

88. What is a "robot"?

A) A machine that can be programmed to do tasks

B) A computer game

C) A movie character

D) A smartphone application

Answer: A) A machine that can be programmed to do tasks

89. What is the purpose of a "training dataset" in AI?

A) To teach the machine how to make decisions based on patterns

B) To store videos

C) To create music

D) To organize photos

Answer: A) To teach the machine how to make decisions based on patterns

90. How does AI help in improving customer service?

A) By providing quick answers through chatbots

B) By sending random offers

C) By sending emails automatically

D) By making calls to customers

Answer: A) By providing quick answers through chatbots

91. What is an example of AI being used in social media?

A) Recommending posts and friends based on your interests

B) Posting random pictures

C) Sending birthday wishes

D) Making phone calls

Answer: A) Recommending posts and friends based on your interests

92. What is the main goal of AI?

A) To make machines intelligent and able to perform tasks that usually require human intelligence

B) To make humans smarter

C) To make toys

D) To organize games

Answer: A) To make machines intelligent and able to perform tasks that usually require human intelligence

93. How does AI help in recognizing objects in pictures?

A) By identifying patterns in the image

B) By guessing the object

C) By drawing the object

D) By using random guesses

Answer: A) By identifying patterns in the image

94. What is an example of AI in transportation?

A) Self-driving cars that make decisions based on surroundings

B) Traffic lights that work automatically

C) Maps that show the best route

D) Cars that stop on their own

Answer: A) Self-driving cars that make decisions based on surroundings

95. What does AI use to improve over time?

A) More data to recognize patterns and make better decisions

B) Human instructions

C) A set of rules to follow

D) Random guessing

Answer: A) More data to recognize patterns and make better decisions

96. What is the role of "data" in AI?

A) Data is used to train the AI to recognize patterns and make predictions

B) Data is used to play games

C) Data is used to make music

D) Data is used to delete files

Answer: A) Data is used to train the AI to recognize patterns and make predictions

97. What is "machine learning"?

A) A method of teaching machines to learn from data and make decisions

B) A way to make machines play games

C) A way to write stories

D) A method to organize books

Answer: A) A method of teaching machines to learn from data and make decisions

98. How does AI help in facial recognition?

A) By identifying facial features like eyes, nose, and mouth

B) By guessing the face

C) By creating random faces

D) By drawing a face

Answer: A) By identifying facial features like eyes, nose, and mouth

99. How does AI help in sorting emails?

A) By identifying spam and organizing important emails

B) By reading emails aloud

C) By sending emails automatically

D) By deleting all emails

Answer: A) By identifying spam and organizing important emails

100. What is "Natural Language Processing" (NLP) in AI?

A) A way for machines to understand and interpret human language

B) A tool to clean data

C) A method to play games

D) A system for organizing information

Answer: A) A way for machines to understand and interpret human language

Open-ended Questions

1. How do you think voice assistants like Siri or Alexa use Machine Learning?
2. How do you think Machine Learning is different from regular computer programming?
3. Why do you think Machine Learning is important?
4. Can you think of an example where a machine might need to "learn" something?
5. What is the coolest thing you've heard about Machine Learning?
6. Why is data important for Machine Learning?
7. If a machine had no data, could it learn? Why or why not?
8. What do you think an algorithm does in Machine Learning?
9. How would you describe a model in Machine Learning?
10. If you were building a model, what would you want it to predict or do?
11. How is supervised learning like being taught by a teacher?
12. What is unsupervised learning, and when do you think it would be useful?
13. Can you think of a situation where supervised learning might be better than unsupervised learning?

14. How do you think a machine figures out patterns in data without being told what to look for?
15. If you were designing a machine to sort books, would you use supervised or unsupervised learning? Why?
16. What do you think is the first step in teaching a machine to learn?
17. Why is it important to clean data before training a machine?
18. What happens when a machine learns from bad or incorrect data?
19. How does a machine know when it has learned enough?
20. How would you explain Machine Learning (ML) to a friend your age?
21. If a machine is making too many mistakes, what would you do to fix it?
22. When you use a search engine, how do you think Machine Learning helps find what you're looking for?
23. Can you think of a game you play that might use Machine Learning?
24. How do you think Machine Learning helps make better recommendations for movies or music?
25. If you were a doctor, how could Machine Learning help you treat patients?
26. Imagine you are creating a Machine Learning system to help farmers. What would you want it to do?
27. How could Machine Learning be used to make transportation safer?
28. If Machine Learning could predict the weather, how could that help people?

29. How do you think Machine Learning can help in creating new medicines?
30. If you were solving the problem of cleaning up plastic in the ocean, how could Machine Learning help?
31. If you were designing a Machine Learning system for a library, what problem would it solve?
32. Imagine a Machine Learning robot in your school. What task would you ask it to do, and why?
33. If Machine Learning could sort photos on your phone, how would you want it to organize them?
34. You are designing a Machine Learning app to help people eat healthier. What features would it have?
35. Imagine Machine Learning was used to design a theme park. What role would it play?
36. A Machine Learning system is trained to recognize dogs in pictures, but it also labels cats as dogs. How would you fix it?
37. Imagine a Machine Learning system predicts your test score. If it gets your score wrong, what might have gone wrong?
38. You're using a voice assistant, but it doesn't understand some of the words you say. How could Machine Learning improve this?
39. If a self-driving car learned from bad data, what could happen? How would you fix it?
40. Imagine you have a Machine Learning app that tracks your study habits. How would it help you improve?

41. If Machine Learning could create a new game, what would you want it to be about?
42. Imagine you could teach a machine anything. What would you want it to learn?
43. If Machine Learning could help you invent a new gadget, what would it be?
44. Can Machine Learning be creative? What kind of artwork do you think it could make?
45. If a machine could learn your favorite hobby, how would it use that knowledge to help you?
46. Do you think machines should always tell people how they made a decision? Why or why not?
47. If Machine Learning made a decision that was unfair, who would be responsible?
48. What are some risks of using Machine Learning, and how can we make it safer?
49. If a Machine Learning app predicted something about you that you didn't agree with, how would you feel?
50. Should there be rules for what Machine Learning can and cannot do? What would some of those rules be?